

Dalhousie University
CSCI 3110 — Design and Analysis of Algorithms I
Fall 2024
Assignment 7

Distributed Tuesday, November 5 2024.

Due 11:59PM, Tuesday, November 19 2024.

Instructions:

1. Before starting to work on the assignment, make sure that you have read and understood policies regarding the assignments of this course, especially the policy regarding collaboration. <https://dal.brightspace.com/d2l/le/content/338900/Home?itemIdentifier=D2L.LE.Content.ContentObject.ModuleC0-4457275>
2. To submit via Crowdmark, upload a **separate** image/PDF file (multiple pages are allowed) to each question. You may resubmit as often as necessary until the due date. A good strategy is to create an initial submission days in advance after you solve some problems, and make resubmissions later. More detailed submission information for Crowdmark can be found at:
<https://crowdmark.com/help/completing-and-submitting-an-assignment/>
3. If you submit a joint assignment, use Crowdmark to create a group and add group members before making submissions. The “What Will Students See” section of <https://crowdmark.com/help/creating-a-group-assignment> shows the steps and screenshots of creating a group and adding members. Note that, “to avoid overwriting submissions”, Crowdmark recommends “only one group member submits all the work”.
4. We encourage you to typeset your solutions using LaTeX. However, you are free to use other software or submit scanned handwritten assignments as long as **they are legible**. We have the right to take marks off for illegible answers.

Questions:

1. [10 marks] Consider the weighted, connected and undirected graph G in Figure 1:
 - (a) [5 marks] Write down the edges of a minimum spanning tree of G constructed using Kruskal’s algorithm, *in the order that they are selected by Kruskal’s algorithm*.
You can use (a, b) to denote an edge between vertices a and b .
You need not actually draw the tree or show your steps.

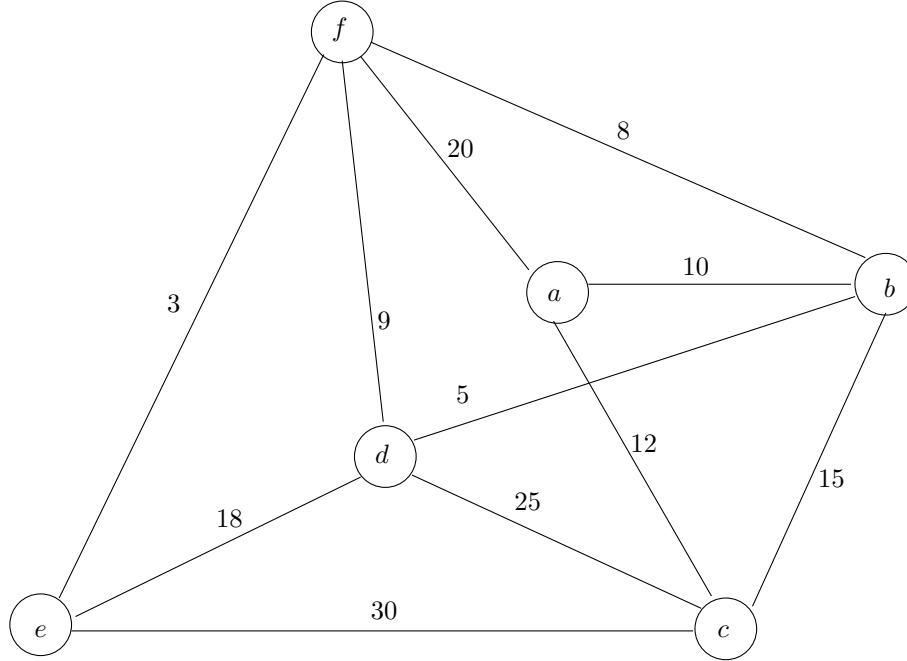


Figure 1: The graph G in question 1.

- (b) [5 marks] Choose vertex a as the *starting vertex* for Prim's algorithm, and write down the edges of a minimum spanning tree of G constructed using Prim's algorithm, *in the order that they are selected by Prim's algorithm*.

Use (a, b) to denote an edge between vertices a and b .

You need not actually draw the tree or show your steps.

2. [10 marks] Consider three strings A , B and C of lengths m , n and $m + n$, respectively. We say that string C is a *mix* of A and B if its set of indices, $\{1, 2, \dots, m + n\}$, can be partitioned into two disjoint subsets $I = \{i_1, i_2, \dots, i_m\}$ (where $i_1 < i_2 < \dots < i_m$) and $J = \{j_1, j_2, \dots, j_n\}$ (where $j_1 < j_2 < \dots < j_n$), such that $C[i_1]C[i_2] \dots C[i_m] = A$ and $C[j_1]C[j_2] \dots C[j_n] = B$.

For example, if $A = \text{alphabet}$, $B = \text{pamphlet}$ and $C = \text{alpaphabmpehtlet}$, then C is a mix of A and B . This is because $C[1]C[2]C[5]C[6]C[7]C[8]C[11]C[13] = A$ and $C[3]C[4]C[9]C[10]C[12]C[14]C[15]C[16] = B$.

Use dynamic programming to design an efficient algorithm to solve the following problem: Given three strings A , B and C of lengths m , n and $m + n$, respectively, return “true” if C is a mix of A and B , and “false” otherwise.

3. [10 marks] Further design an efficient algorithm to compute the subsets I and J as defined in question 2 if C is a mix of A and B . Since $J = \{1, 2, \dots, m + n\} \setminus I$, it is

sufficient if your algorithm returns the elements in I only. If C is not a mix of A and B , return the set $\{-1\}$.